

Remote Control Unit (RCU)

Hardware and Software Reference

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KJO Liquid Rocket Motor Control System

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This document describes the hardware, software architecture, operator interface, and control flow of the Remote Control Unit. The RCU is the sole operator interface to the KJO liquid rocket motor control system.

Contents

1	Hardware Overview	2
1.1	Component Summary	2
1.2	Pin Assignments	2
1.3	Hardware Block Diagram	2
2	Software Architecture	2
2.1	Module Summary	3
2.2	Main Loop Control Flow	3
3	Operator Interface (GUI)	3
3.1	Button Layout and Behaviour	4
3.2	Multi-Source Ignition	4
3.3	Fill Progress Popup	5
3.4	Valve Status Indicators	5
3.5	Status Panels	5
4	Engine Start Confirmation Dialog	5
5	Pressure Warning	6
6	Radio Command Dispatch	6
6.1	Command Reference	6
7	SD Logging	7
8	Parameters & Flags	7
9	Version History	9

1 Hardware Overview

1.1 Component Summary

Component	Part	Interface
Microcontroller	Adafruit Feather M0 (ATSAMD21G18A, 48 MHz)	—
Display	Adafruit 3.5" TFT V2 (HX8357D, 480×320)	SPI
Touchscreen	TSC2007 resistive controller	I ² C (0x48)
LoRa Radio	HopeRF RFM95W (Semtech SX1276), 915 MHz	SPI
Real-Time Clock	DS3231 (provides temperature also)	I ² C
SD Card	On Adafruit 3.5" TFT shield	SPI
Battery	3.7 V LiPo; voltage divider on A7	ADC A7

1.2 Pin Assignments

Signal	Pin	Notes
SCREEN_CS	11	TFT chip select
SCREEN_DC	10	TFT data/command
SD_CS	5	SD card chip select
STMPE_CS	6	Touch pen-down IRQ (TOUCH_IRQ aliases this pin). Legacy name from an earlier STMPE610-era touch controller; the current TSC2007 is pure I ² C and never uses this pin as an SPI chip-select.
RFM95_CS	8	Radio chip select
RFM95_RST	4	Radio reset
RFM95_INT	3	Radio IRQ
BAT_SENSE	A7	Battery voltage (2:1 divider)

1.3 Hardware Block Diagram

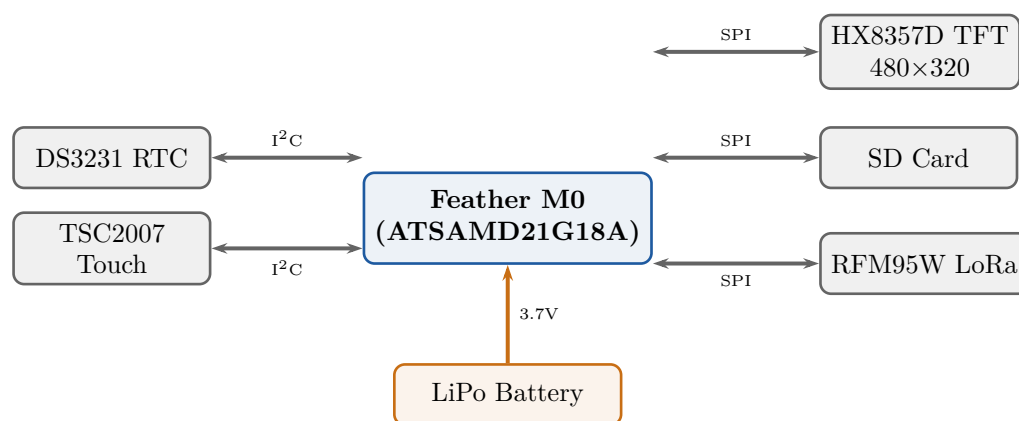


Figure 1: RCU hardware block diagram.

2 Software Architecture

2.1 Module Summary

Module	Location	Responsibility
main.cpp	local	Setup, loop, command dispatch, GUI logic
KJO_RCU_Display	shared	TFT GUI: buttons, valve symbols, status panels
KJO_Radio	shared	Radio pin constants, frequency, power
KJO_Command_Defs	shared	Command codes, packet struct definitions
KJO_System_Config	shared	Version string, platform identifier
KJO_Status	shared	Tagged Serial/SD logging (<code>Log_Message()</code>); RCU's single-line message panel (<code>Display_Message()</code>) does not use the scrolling <code>KJO_Status_Display</code> module the other three boards share

2.2 Main Loop Control Flow

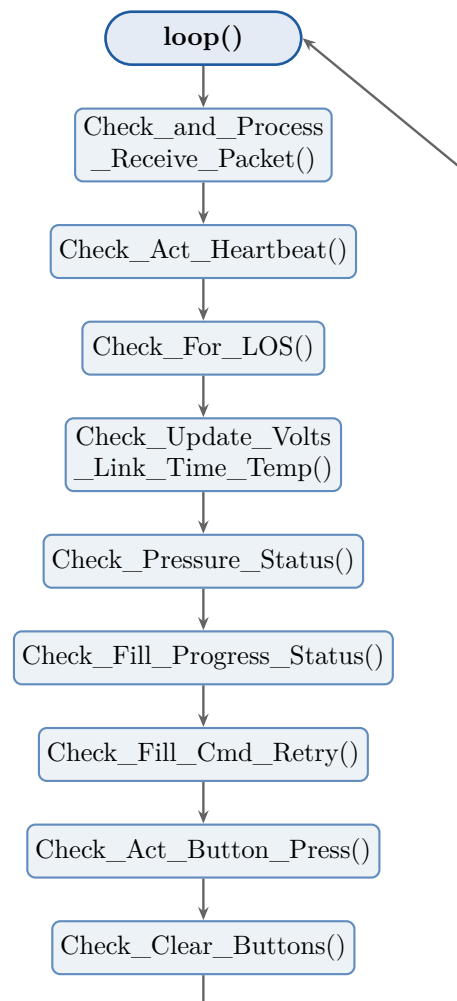


Figure 2: RCU main loop control flow.

3 Operator Interface (GUI)

3.1 Button Layout and Behaviour

Nine buttons are arranged in three rows plus one relocated button, reflecting the multi-source-ignition redesign (see §3.2):

- **Row 1** (3 equal columns): Fill, Dump, Status.
- **Row 2** (2 columns, original button geometry): Arm/Disarm, Launch.
- **Row 3**: three small ignition-source-select squares (RCU / LCO / AS), left-aligned under Arm/Disarm.
- **Abort**: relocated off the button grid entirely – below the Fuel valve indicator, in the lower-right of the screen. Sends the same **RESET** as an earlier “Reset” button used to send; there is no functional distinction between “Reset” and “Abort” in this system, only the relocation and renaming.

Button	Behaviour
Fill	Every press sends BEGIN_FILL and opens the fill-progress popup (see §3.3) – EMU’s own fill-session state decides fresh-start vs. resume, so RCU doesn’t need to track which. Ignored once the button shows complete (green); RESET is required before filling again.
Dump	Latching toggle . First press → DUMP (button stays highlighted). Second press → STOP_DUMP (button clears).
Status	Sends RCU_STATUS ; momentary highlight, auto-clears.
Arm/Disarm	Tap-to-toggle . Label always shows the action a press will take. Sends ARM_LAUNCH (param 1 to arm, 0 to disarm) and shows a pending state until EMU acknowledges; the label/highlight only commits to Armed/Disarmed on that ack (reverts on failure, e.g. no ignition source selected).
Launch	Guarded by confirmation dialog . Press opens a modal CONFIRM / CANCEL overlay; only CONFIRM attempts START_ENGINE , and only if still armed for the RCU source at that moment – see §4.
RCU / LCO / AS	Ignition-source-select squares. Tap sends SET_IGNITION_SOURCE with the corresponding param (0/1/2); shows pending until EMU acknowledges. Rejected by EMU while armed – disarm first to change source.
Abort	Immediate . Sends RESET ; clears all latched button states (Fill, Dump) and closes the fill popup if open; brief red flash for visual confirmation.

Note

While armed (**ARM_LAUNCH** acknowledged), only Abort, Arm/Disarm, and Launch (and only if the RCU source is currently armed) respond to touch – every other button press is silently ignored until disarmed.

3.2 Multi-Source Ignition

The engine can be ignited from one of three sources, selected on EMU via **SET_IGNITION_SOURCE** (param 0=RCU, 1=LCO, 2=Airstart) and rejected by EMU while armed – the operator must disarm first to change source. Once a source is selected, **ARM_LAUNCH** arms or disarms it. Only the RCU source’s **START_ENGINE** is ever sent from this touchscreen; LCO and Airstart trigger the ignition sequence through their own independent paths on EMU/GSEMU, with RCU’s role limited to selecting and arming/disarming the source and observing the unprompted

ENGINE_START_ARMED notification EMU sends the instant any source arms the sequence.

3.3 Fill Progress Popup

Pressing Fill opens a modal popup that intercepts all touch except Abort (which still passes through, so an operator can always abort out of a fill in progress). While open:

- RCU polls **QUERY_FILL_PROGRESS** every **FILL_PROGRESS_STATUS_INTERVAL** (1.5 s) – see §10 Parameters & Flags – updating the Fill button’s colour and, if the popup is still open, its displayed numbers.
- A toggle button reads **Pause** or **Resume** depending on the last known fill state, sending **STOP_FILL** or **BEGIN_FILL** respectively.
- A close control dismisses the popup without changing fill state; the fill continues in the background and the Fill button keeps reflecting its progress.
- If EMU responds **RELAY_BUSY** to a Fill/Pause/Resume attempt, RCU automatically retries once after a short delay (**FILL_CMD_RETRY_DELAY_MS**).

3.4 Valve Status Indicators

Three valve symbols occupy the right portion of the screen: Fill, N₂O (Ox), and Fuel. Each symbol uses a circle-and-bar motif: a vertical bar on a green disc indicates open; a horizontal bar on a red disc indicates closed. State is updated from the **HEARTBEAT** response packet. There is no dedicated Dump indicator — the **DUMP** command opens the Ox valve (vents pressurant through the chamber), so its state is reflected on the Ox indicator.

When the tank pressure exceeds **PRESSURE_IN_SYSTEM_THRESHOLD** (polled via **READ_TANK_PRESSURE**, *not* carried in the heartbeat packet), the background of the valve panel turns **red** as a visual pressure warning (see §5).

3.5 Status Panels

- **Battery voltages:** RCU voltage (measured locally on A7); EMU voltage (received in heartbeat).
- **Link status:** **LINK_OK** / **LINK_MARGINAL** / **LINK_DOWN**, with current RSSI value.
- **Time and temperature:** Time from DS3231; temperature converted to °F.
- **QR/CAN status:** a persistent, heartbeat-driven readout (**Update_QR_Status_Display()**) showing umbilical connection state and GSEMU/LCMU CAN reachability, all three carried in every **HEARTBEAT** response.
- **Message panel:** a single-line status area (**Display_Message()**) showing the most recent status message – *not* a scrolling log, and it never displays raw TX/RX traffic (that goes to the SD log only, see §9).

4 Engine Start Confirmation Dialog

Pressing the **Launch** button does *not* immediately send **START_ENGINE**. Instead, a modal overlay is drawn over the main screen with two large buttons:

- **CONFIRM** (green) — attempts **START_ENGINE**, but *only* if still armed for the RCU source at the moment of touch. If disarmed or re-armed for a different source in the meantime

(e.g. by another operator, or a race with an in-flight **ARM_LAUNCH/SET_IGNITION_SOURCE** ack), the command is never sent and the message panel shows “Launch cancelled – not armed.” instead.

- **CANCEL** (red) — dismisses the overlay without transmitting any command.

Touching anywhere outside both buttons while the overlay is open also dismisses it (silently, without the “START cancelled.” message **CANCEL** shows) – there is no way for the overlay to remain stuck open, and an accidental touch elsewhere never initiates the ignition sequence.

5 Pressure Warning

`Check_Pressure_Status()` is called every loop iteration, but only actively polls **READ_TANK_PRESSURE** (every **PRESSURE_STATUS_INTERVAL**, 2 s) while a fill is active *or* the warning is already active. Tank pressure is *not* carried in the heartbeat packet at all – this poll is RCU’s only source for it.

Warning

Because polling only starts once a fill is active or the warning is already latched, a pressure rise that occurs outside those two windows (e.g. between fill sessions) is not detected by RCU until the next fill starts or another mechanism raises it. This is a real monitoring gap in the current design, not a documentation nuance.

Each **READ_TANK_PRESSURE** response is compared against **PRESSURE_IN_SYSTEM_THRESHOLD**. When pressure exceeds it:

- The valve symbol panel background is redrawn in **red** (suppressed while the fill popup covers that region; repainted from the cached flag when the popup closes).
- A “PRESSURISED” warning label is displayed in the status area.

When a subsequent reading returns below the threshold, the panel background returns to its normal colour and the warning label is cleared.

This visual warning mirrors the EMU-side audible pressure alarm (continuous buzzer on the EMU) so the operator at the RCU can see that the propellant system is under pressure, but only while the poll described above is actually running.

6 Radio Command Dispatch

`RCU_Execute_Command(command_ID, p1)` builds and transmits a command packet. **HEARTBEAT** uses the `Heartbeat_Command_Packet` structure – just the 1-byte command code, *not* RCU voltage (a prior design sent RCU voltage here; it was removed as wasted payload the EMU had no use for). All other commands use `Command_Payload` (4 bytes, includes a **short** parameter).

Responses are parsed in `Check_and_Process_Receive_Packet()`. The **HEARTBEAT** response is routed to `Process_Heartbeat_Response()`, which updates EMU voltage, the three valve symbols (Fill/Ox/Fuel), and the QR/GSEMU/LCMU status readout – all carried in the same packet.

6.1 Command Reference

Command	Sent by	Notes
HEARTBEAT	Automatic (every <code>heartbeat_interval</code> , 5 s, when no other traffic has occurred)	Keeps the link alive; EMU returns voltage, valve positions, and QR/GSEMU/LCMU status
RESET	Abort button	Immediate abort; clears all latched button states and closes the fill popup if open
BEGIN_FILL	Fill button (every press); also the fill popup's Resume toggle, and an automatic <code>RELAY_BUSY</code> retry	Starts/resumes propellant fill; EMU starts pressure recording
STOP_FILL	Fill popup's Pause toggle; automatic <code>RELAY_BUSY</code> retry	Pauses propellant fill; recording continues until ambient
QUERY_FILL_PROGRESS	Automatic, every <code>FILL_PROGRESS_STATUS_INTERVAL</code> while a fill is active	Drives the Fill button's colour and the popup's displayed numbers
READ_TANK_PRESSURE	Automatic, every <code>PRESSURE_STATUS_INTERVAL</code> , while a fill is active or the pressure warning is already active	Drives the pressure-warning display (see §5) – not a continuous poll
DUMP	Dump button (1st press)	Opens Ox valve; vents pressurant through the chamber (no dedicated Dump valve)
STOP_DUMP	Dump button (2nd press)	Closes Ox valve
SET_IGNITION_SOURCE	RCU/LCO/AS source buttons	Selects the active ignition source (param 0/1/2); rejected by EMU while armed
ARM_LAUNCH	Arm/Disarm button	Arms or disarms the currently-selected source (param 1/0); renamed from <code>ENABLE_LCO_CONTROL</code> – see §11 Version History
START_ENGINE	Launch + CONFIRM (only if still armed for the RCU source)	Arms the 10-step ignition sequence on EMU
RCU_STATUS	Status button	Requests EMU status summary

7 SD Logging

Log files are stored in the root of the SD card as `Log_N.txt` (N is auto-incremented to avoid overwriting existing files). Each entry uses the shared `Log_Message()` format ("`MM/DD/YYYY HH:MM:SS.mmm [TAG] message`"), timestamped from the DS3231 (falling back to a raw `millis()` count if the RTC never initialized). TX events use `TXR`; RX events use `RXR`; other tags cover boot/status/error events, e.g.:

```
09/09/2026 10:15:03.142 [TXR] BEGIN_FILL (1) | param: 0
09/09/2026 10:15:03.318 [RXR] BEGIN_FILL (1) | status: OK | r_val: 0 |
RSSI: -72
```

8 Parameters & Flags

This section collects the tunable operational parameters and feature flags compiled into this firmware image, together with the source file where each is defined. It is meant as a single reference point when tracing or changing a timing, threshold, or calibration value. Pin/channel wiring assignments, LoRa command opcodes, and packet byte layouts are documented separately (the Hardware Overview pin map and the Command Reference sections above) and are not

repeated here. Parameters marked (*shared*) are defined in `KJO_Shared_Libraries` and must agree with the same constant used on the other board(s) that consume it.

Radio Communication

Parameter	Value	Description	Source File
<code>RCS_CENTER_FREQUENCY</code>	917.117 MHz	LoRa carrier frequency; must match on both RCU and EMU radios to communicate.	<code>KJO_Radio.h</code> (shared)
<code>TRANSMIT_POWER</code>	5 dBm	RFM95 transmit power (PA_BOOST pin), valid range 5–23 dBm.	<code>KJO_Radio.h</code> (shared)
<code>LINK_MARGINAL_THRESHOLD</code>	−90 dBm	RSSI floor below which the link status indicator shows MARGINAL (yellow) instead of OK (green).	<code>KJO_Radio.h</code> (shared)
<code>MAX_PAYLOAD_LENGTH</code>	64 bytes	Size of the LoRa TX/RX payload buffer; sized with headroom above the largest packet (<code>Extended_Heartbeat_Response_Packet</code> , 39 bytes).	<code>KJO_Command_Defs.h</code> (shared)
<code>heartbeat_interval</code>	5000 ms	Interval at which a HEARTBEAT command is sent to the EMU when no other traffic has occurred, to keep the radio link alive.	<code>main.cpp</code>
<code>LOS_timeout_interval</code>	30000 ms	Time since last received packet after which the link is declared LOST (<code>LINK_DOWN</code>) and RSSI/status reset.	<code>main.cpp</code>
<code>FILL_CMD_RETRY_DELAY_MS</code>	250 ms	Delay before firing the single automatic <code>BEGIN_FILL/STOP_FILL</code> retry after EMU responds <code>RELAY_BUSY</code> .	<code>main.cpp</code>

GUI / Display Timing

Parameter	Value	Description	Source File
<code>volt_RSSI_update_interval</code>	1000 ms	Refresh interval for the battery voltage, link status, time, and temperature display fields.	<code>main.cpp</code>
<code>show_button_duration</code>	1000 ms	How long a momentary button stays highlighted (pressed color) before reverting, for non-latching buttons.	<code>main.cpp</code>

Fill Progress Polling

Parameter	Value	Description	Source File
<code>FILL_PROGRESS_STATUS_INTERVAL</code>	1500 ms	Poll interval for <code>QUERY_FILL_PROGRESS</code> while a fill session is active, driving the Fill button color and popup progress bar.	<code>main.cpp</code>
<code>PRESSURE_STATUS_INTERVAL</code>	2000 ms	Poll interval for <code>READ_TANK_PRESSURE</code> while a fill is active or the pressure warning is active.	<code>main.cpp</code>

Pressure Monitoring

Parameter	Value	Description	Source File
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PRESSURE_IN_SYSTEM_THRESHOLD	10 PSI	Tank pressure above which the RCU displays the “system pressurized” warning (valve panel turns red).	KJO_RCU_Display.h (shared)
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SD Logging

Parameter	Value	Description	Source File
LOG_FILE_NAME_BASE	"Log_"	Base filename prefix for SD-card log files; combined with an auto-incrementing index (Log_0.txt, Log_1.txt, ...) to find the next unused file at boot.	main.cpp

Battery Voltage Calibration (RCU Local ADC)

Parameter	Value	Description	Source File
RCU_Batt_Info.AD_res	12 bits	ADC resolution used when sampling the RCU’s own battery voltage divider on pin A7.	main.cpp
RCU_Batt_Info.AD_samples	4	Number of ADC samples averaged per RCU battery voltage reading (noise reduction).	main.cpp

9 Version History

Version	Date	Notes
0.1	March 2026	Initial release.
0.2	March 2026	Button table updated: FILL and DUMP are latching toggles; Reset is immediate ABORT and clears all latched states; Enable is hold-to-enable LCO (lift = disable); Launch guarded by confirmation dialog. Main loop diagram updated to include Check_Pressure_Status(). New §4 Engine Start Confirmation Dialog. New §5 Pressure Warning (valve panel background turns red).
0.3	March 2026	SD log files moved to SD card root (Log_N.txt); subdirectory path constant (LOG_FILE_FOLDER) removed from main.cpp.
0.4	July 2026	Dump valve physically removed from the vehicle. DUMP/STOP_DUMP retained (still a latching toggle) but now open/close the Ox valve instead of a dedicated Dump valve; the separate Dump valve status indicator was removed and its state folded into the Ox indicator.
0.5	September 2026	Add new §8 Parameters & Flags reference section: tunable timing, threshold, and calibration constants across the radio, GUI, fill-progress polling, and pressure-monitoring subsystems, with source-file attribution for each.

0.6	September 2026	Comprehensive doc-vs-code audit – this document had fallen well behind two undocumented redesigns already live in the firmware (multi-source ignition, ~2026-07-23; fill-progress popup, ~2026-08-23), neither reflected here until now. Rebuilt §3 Operator Interface around the actual 9-button layout (Fill/Dump/Status, Arm-Disarm/Launch, RCU/LCO/AS source select, a relocated Abort) – replacing a stale 6-button Fill/Dump/Reset/Status/Enable/Launch grid; removed the nonexistent ENABLE_LCO_CONTROL command (renamed ARM_LAUNCH months ago) and documented SET_IGNITION_SOURCE/ ARM_LAUNCH and the armed-state lockout (new §3.2 Multi-Source Ignition); documented the fill-progress popup (new §3.4) and corrected the Fill button’s actual behavior (not a toggle); corrected the Launch confirmation dialog (CONFIRM has a real precondition; touching outside the overlay also silently dismisses it); corrected the Pressure Warning section (tank pressure is never in the heartbeat packet, and polling is <i>not</i> continuous – a real monitoring gap, not just a doc error); rebuilt the Command Reference table (removed the nonexistent command, added 4 missing ones, fixed two mislabeled triggers); corrected the Heartbeat_Command_Packet size (1 byte, not 6 – RCU voltage was removed from it); fixed §4 Main Loop Control Flow (2 missing steps); fixed the SD Logging example (real timestamp format and tags); corrected the STMPE_CS pin description (legacy name; it’s a touch IRQ, not an SPI chip-select on the current TSC2007 hardware); added KJO_Status to the Module Summary.
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